

DSA / Algorithms (Abdul Bari) - Day 44

45-day plan mapped to the Algorithms playlist

Rabin-Karp + AVL Trees

Rabin-Karp hashing match and AVL tree rotations.

Watch these Abdul Bari videos

- > 9.2 Rabin-Karp String Matching Algorithm
- > 10.1 AVL Tree - Insertion and Rotations

Key concepts

- Rabin-Karp uses a rolling hash to compare windows.
- AVL keeps a BST balanced with rotations.
- Balance factor in $[-1,1]$.

Key idea

AVL: LL, RR, LR, RL rotations restore balance after insert.

Tasks (1.5h)

- ☐ Watch both videos
- ☐ Code a rolling hash
- ☐ Do AVL rotations

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0